

APPROVAL OF REQUIREMENTS FOR PROJECT AURORA

INTRODUCTION

1. Project AURORA seeks approval to conduct a research and technology study into the use of advanced wireless sensing for non-contact detection and monitoring in indoor environments.

The project will be conducted as a research collaboration with an Institute of Higher Learning (IHL). The project team will work with university researchers to investigate the feasibility and performance of the proposed sensing technology under representative operating conditions.

2. The study is intended to evaluate the potential of the emerging technology, establish its technical limitations, and determine whether it is sufficiently mature for further development or operational application.

JUSTIFICATIONS

3. Existing sensing solutions require dedicated sensors to be installed at specific locations and may have limitations in coverage and deployment flexibility.

Recent developments in wireless sensing present an opportunity to investigate whether existing wireless signals can be used to detect changes in the surrounding environment without requiring additional dedicated sensing infrastructure.

However, the performance and feasibility of this emerging technology have not been sufficiently validated for the intended operational context. A research study is therefore required before committing resources to further development or implementation.

Project AURORA will allow the organisation to investigate the technology, conduct controlled experimentation, and obtain research evidence to support subsequent decisions on whether the technology should be pursued.

SCOPE OF WORK

4. The Project Manager will work with the appointed Institute of Higher Learning to define and conduct the research Scope of Work.

The research activities will include:

- Review of existing wireless sensing technologies and research.
- Definition of research objectives and performance measures.
- Investigation of suitable sensing techniques and technical approaches.
- Design and conduct of controlled experiments.
- Collection and analysis of experimental data.
- Evaluation of technology performance under different environmental conditions.
- Identification of technical limitations and implementation challenges.
- Demonstration of selected research outcomes.
- Assessment of the feasibility of further technology development or operational application.

The IHL research team will document the experimental methodology, research findings, technology evaluation, and recommendations in a final research report.

The expected research outcomes include an assessment of the technical feasibility of the proposed technology, its potential applications, identified limitations, and recommendations for subsequent research or development.

IMPLEMENTATION DETAILS

5. Following project initiation and creation, Project AURORA will be established and managed through the organisation's online project planning and management system.

Front-end project support will assist with the establishment of the project and coordinate the necessary project staffing arrangements.

6. The Project Manager will work with the appointed IHL and relevant internal stakeholders to finalise the research objectives, Scope of Work, research activities, expected outcomes, project milestones, and research deliverables.

7. The research will be conducted through a series of technology investigations and experiments. Experimental findings will be reviewed periodically with the project team to assess research progress and determine whether additional investigations are required.

For selected experiments, the research equipment may require connectivity to the organisation's intranet to support controlled data collection and technology evaluation.

Prior to establishing any required intranet connectivity, the project team will conduct the necessary cybersecurity assessment. A Cyber Threat Risk Assessment (TRA) will be performed to identify and assess cybersecurity threats and risks associated with the research equipment and proposed system connectivity.

8. The TRA and proposed connectivity arrangements will be submitted to the appropriate approving authority for review and endorsement before connection to the intranet is permitted.

Research findings will be reviewed at the completion of the project to determine whether the technology demonstrates sufficient potential to justify further research, technology development, or implementation.

COST AND SCHEDULE

9. The total project cost is estimated to be \$0.195M.

Estimated Cost

Item	Estimated Cost
IHL research manpower	\$120,000
Research equipment and materials	\$35,000
Experimental activities	\$25,000
Technology evaluation and demonstration	\$15,000
Total	\$195,000

Indicative Schedule

Activity	Timeline
Project initiation and research planning	Month 1
Finalisation of research Scope of Work	Month 1
Cyber Threat Risk Assessment and endorsement	Months 1–2
Technology investigation and experimental setup	Months 2–3
Experimental research and data collection	Months 3–6

Technology evaluation and analysis	Months 6–7
Research demonstration	Month 8
Final research report and recommendations	Month 9

The detailed experimental activities may be refined during the project based on research findings and the results of preceding technology investigations.

RECOMMENDATION

10. It is recommended that the Director approve Project AURORA at an estimated total cost of \$0.195M and allow the project to proceed through the R&T QMS process for the conduct of the proposed research and technology activities.